

pyran-4-ylamino)benzamide (1200 mg/m²/day, Schedule A). The patient exhibited a partial response that is ongoing (patient currently in cycle 9).

[0701] A 63 year old male, ROS1 rearrangement, s/p 3 cycles of chemotherapy, achieved partial response after 1 cycle of N-[5-(3,5-difluorobenzyl)-1H-indazol-3-yl]-4-(4-methyl-piperazin-1-yl)-2-(tetrahydro-2H-pyran-4-ylamino)benzamide (400 mg/m²/day, Schedule C). Partial response is ongoing (patient currently in cycle 3).

[0702] Colorectal Cancer (CRC) (N=1).

[0703] A 75 year old female with metastatic CRC, TrkA rearrangement, s/p 3 cycles of chemotherapy, cetuximab, achieved partial response after 1 cycle of N-[5-(3,5-difluorobenzyl)-1H-indazol-3-yl]-4-(4-methyl-piperazin-1-yl)-2-(tetrahydro-2H-pyran-4-ylamino)benzamide (1600 mg/m², Schedule A). Partial response lasted for 4 cycles before progressive disease.

[0704] Neuroblastoma (NB) (N=1).

[0705] A 22 year old female with metastatic NB, ALK mutation, s/p 4 cycles of chemotherapy, achieved partial response after 12 cycles of N-[5-(3,5-difluorobenzyl)-1H-indazol-3-yl]-4-(4-methyl-piperazin-1-yl)-2-(tetrahydro-2H-pyran-4-ylamino)benzamide (started at 200 mg/m²/day and had subsequent dose escalations to 1200 mg/m²/day, Schedule A). Partial response is ongoing (the patient is currently in cycle 21).

[0706] A total of 25 patients were evaluated across 3 different dosing schedules in this trial of N-[5-(3,5-difluorobenzyl)-1H-indazol-3-yl]-4-(4-methyl-piperazin-1-yl)-2-(tetrahydro-2H-pyran-4-ylamino)benzamide.

[0707] There was a complete response in a patient with non-small cell lung cancer (NSCLC) with a ROS1 molecular alteration. There were 5 partial responses (3 of these durable >9 cycles) across multiple tumor types (NSCLC, CRC and NB) in patients with TrkA, ROS1, and ALK molecular alterations; moreover, there were 2 patients with prolonged stable disease (11+ cycles) with NSCLC and pancreatic adenocarcinoma.

[0708] N-[5-(3,5-difluorobenzyl)-1H-indazol-3-yl]-4-(4-methyl-piperazin-1-yl)-2-(tetrahydro-2H-pyran-4-ylamino)benzamide was well tolerated in all dosing schedules tested, including intermittent and daily continuous administration.

[0709] Most common AEs (mainly grade 1-2) considered possibly drug-related were nausea, paresthesia, vomiting, diarrhea, asthenia, myalgia, arthralgia, and dysgeusia.

[0710] Only 1 possibly drug-related Grade >3 AE was observed: asthenia (Grade 3 in one patient).

[0711] No drug limiting toxicities, study discontinuations due to adverse events, or drug-related serious adverse events were reported.

[0712] Schedule A was terminated at 1600 mg/m²/day due to a plateau of N-[5-(3,5-difluorobenzyl)-1H-indazol-3-yl]-4-(4-methyl-piperazin-1-yl)-2-(tetrahydro-2H-pyran-4-ylamino)benzamide exposure and the data were presented previously.

[0713] N-[5-(3,5-difluorobenzyl)-1H-indazol-3-yl]-4-(4-methyl-piperazin-1-yl)-2-(tetrahydro-2H-pyran-4-ylamino)benzamide was well tolerated in all dosing schedules tested (both on an intermittent and daily continuous basis). Most AEs <Grade 2.

[0714] Only 1 possibly drug-related Grade >3 AE has been observed: asthenia (1200 mg/m², Schedule A), that subsided with dose reduction to 800 mg/m²/day.

[0715] No DLTs, study discontinuations due to AEs, or drug-related SAEs were reported in any dosing schedule.

[0716] Exposure to N-[5-(3,5-difluorobenzyl)-1H-indazol-3-yl]-4-(4-methyl-piperazin-1-yl)-2-(tetrahydro-2H-pyran-4-ylamino)benzamide increased in an approximate dose proportional manner up to doses of 800 mg/m² with mean plasma half-life of 17 to 44 hours.

[0717] Responders tended to have higher exposure than non-responders throughout the entire dosing cycle in Schedule A.

[0718] The term “comprising” as used herein is synonymous with “including,” “containing,” or “characterized by,” and is inclusive or open-ended and does not exclude additional, unrecited elements or method steps.

[0719] The term “wild-type” as used herein refers to the protein, nucleic acid sequence, allele, locus, or activity level of a protein in a disease-free cell of a healthy individual with respect to a disease in question. For example, a wild-type ROS1 activity level corresponds to an activity level of ROS1 in a healthy cell of an individual lacking a ROS1-related disease or pre-disease condition.

[0720] All numbers expressing quantities of ingredients, reaction conditions, and so forth used in the specification are to be understood as being modified in all instances by the term “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth herein are approximations that may vary depending upon the desired properties sought to be obtained. At the very least, and not as an attempt to limit the application of the doctrine of equivalents to the scope of any claims in any application claiming priority to the present application, each numerical parameter should be construed in light of the number of significant digits and ordinary rounding approaches.

[0721] The above description discloses several methods and materials of the present invention. This invention is susceptible to modifications in the methods and materials, as well as alterations in the fabrication methods and equipment. Such modifications will become apparent to those skilled in the art from a consideration of this disclosure or practice of the invention disclosed herein. Consequently, it is not intended that this invention be limited to the specific embodiments disclosed herein, but that it cover all modifications and alternatives coming within the true scope and spirit of the invention.

[0722] All references cited herein, including but not limited to published and unpublished applications, patents, and literature references, are incorporated herein by reference in their entirety and are hereby made a part of this specification. To the extent publications and patents or patent applications incorporated by reference contradict the disclosure contained in the specification, the specification is intended to supersede and/or take precedence over any such contradictory material.

What is claimed is:

1. A method of treating cancer in a patient, wherein said patient is known to possess at least one genetic alteration in at least one target gene selected from ROS1, NTRK1, NTRK2, and NTRK3, comprising administering to said cancer patient a therapeutically effective amount of a compound selected from N-[5-(3,5-difluorobenzyl)-1H-indazol-3-yl]-4-(piperazin-1-yl)-2-(tetrahydro-2H-pyran-4-ylamino)benzamide, N-[5-(3,5-difluorobenzyl)-1H-indazol-3-yl]-4-(4-methyl-piperazin-1-yl)-2-(tetrahydro-2H-pyran-4-ylamino)benzamide, and N-[5-(3,5-difluoro-benzenesulfonyl)-1H-indazol-

3-yl]-2-((R)-2-methoxy-1-methyl-ethylamino)-4-(4-methyl-piperazin-1-yl)benzamide, or a pharmaceutically acceptable salt thereof.

2. A method according to claim 1, wherein said at least one target gene is ROS1.

3. A method according to claim 1, wherein said at least one target gene is selected from at least one of NTRK1, NTRK2, and NTRK3.

4. A method according to claim 1, wherein said compound is N-[5-(3,5-difluorobenzyl)-1H-indazol-3-yl]-4-(piperazin-1-yl)-2-(tetrahydro-2H-pyran-4-ylamino)benzamide, or a pharmaceutically acceptable salt thereof.

5. A method according to claim 4, wherein said at least one target gene is ROS1.

6. A method according to claim 4, wherein said at least one target gene is selected from at least one of NTRK1, NTRK2, and NTRK3.

7. A method according to claim 1, wherein said compound is N-[5-(3,5-difluorobenzyl)-1H-indazol-3-yl]-4-(4-methyl-piperazin-1-yl)-2-(tetrahydro-2H-pyran-4-ylamino)benzamide, or a pharmaceutically acceptable salt thereof.

8. A method according to claim 7, wherein said at least one target gene is ROS1.

9. A method according to claim 7, wherein said at least one target gene is selected from at least one of NTRK1, NTRK2, and NTRK3.

10. A method according to claim 1, wherein said compound is N-[5-(3,5-difluoro-benzenesulfonyl)-1H-indazol-3-yl]-2-((R)-2-methoxy-1-methyl-ethylamino)-4-(4-methyl-piperazin-1-yl)benzamide, or a pharmaceutically acceptable salt thereof.

11. A method according to claim 10, wherein said at least one target gene is ROS1.

12. A method according to claim 10, wherein said at least one target gene is selected from at least one of NTRK1, NTRK2, and NTRK3.

13. A method according to claim 1, wherein said cancer is selected from non-small cell lung cancer, papillary thyroid cancer, neuroblastoma, pancreatic cancer and colorectal cancer.

14. A method according to claim 1, wherein said compound is N-[5-(3,5-difluorobenzyl)-1H-indazol-3-yl]-4-(piperazin-1-yl)-2-(tetrahydro-2H-pyran-4-ylamino)benzamide, or a pharmaceutically acceptable salt thereof, and said cancer is selected from non-small cell lung cancer, papillary thyroid cancer, neuroblastoma, pancreatic cancer and colorectal cancer.

15. A method according to claim 14, wherein said cancer is selected from non-small cell lung cancer, neuroblastoma, and colorectal cancer.

16. A method according to claim 1, wherein said compound is N-[5-(3,5-difluorobenzyl)-1H-indazol-3-yl]-4-(4-methyl-piperazin-1-yl)-2-(tetrahydro-2H-pyran-4-ylamino)benzamide, or a pharmaceutically acceptable salt thereof, and said cancer is selected from non-small cell lung cancer, papillary thyroid cancer, neuroblastoma, pancreatic cancer and colorectal cancer.

17. A method according to claim 16, wherein said cancer is selected from non-small cell lung cancer, neuroblastoma, and colorectal cancer.

18. A method according to claim 1, wherein said compound is N-[5-(3,5-difluoro-benzenesulfonyl)-1H-indazol-3-yl]-2-((R)-2-methoxy-1-methyl-ethylamino)-4-(4-methyl-piperazin-1-yl)benzamide, or a pharmaceutically acceptable salt thereof, and said cancer is selected from non-small cell lung cancer, papillary thyroid cancer, neuroblastoma, pancreatic cancer and colorectal cancer.

19. A method according to claim 18, wherein said cancer is selected from non-small cell lung cancer, neuroblastoma, and colorectal cancer.

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